

A three-dimensional fluid source placed at the origin induces a radial motion given by $v_r = Q(t)/(4\pi r^2)$, with $v_\theta = v_\phi = 0$. Obtain:

1. Streamlines, trajectories and paths.
2. Equation for the fluid surface initially located at $r = R$.
3. Equation for the fluid line that corresponds at the initial instant with a circle of radius R intersecting the origin.
4. Streamtube that intersects the circle defined by $r = R_1$ and $\theta = \theta_1$.
5. The velocity components in cartesian coordinates, as well as the corresponding streamlines and trajectories.
6. Equations for the streamlines and trajectories for an observer moving with velocity U .

1) $\frac{dr}{v_r} = \frac{r d\theta}{v_\theta} = \frac{r \sin\theta d\phi}{v_\phi} \Rightarrow d\theta = d\phi = 0$ $\theta = \theta_0$
 $\phi = \phi_0$ STREAMLINES ARE STRAIGHT LINES INTERSECTING THE ORIGIN

$\frac{dr}{dt} = \frac{Q(t)}{4\pi r^2}$ $\frac{4\pi}{3}(r^3 - r_0^3) = \int_0^t Q dt$

$\frac{r d\theta}{dt} = 0 \rightarrow \theta = \theta_0$
 $\frac{r \sin\theta d\phi}{dt} = 0 \rightarrow \phi = \phi_0$ } THESE TWO ERS. DEFINE THE PATH LINES

2)  $\left. \begin{matrix} r_0 = R \\ \theta = \alpha \\ \phi = \beta \end{matrix} \right\} \rightarrow \frac{4\pi}{3}(r^3 - R^3) = \int_0^t Q dt$

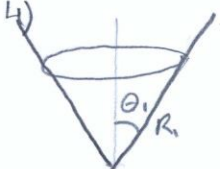
$\theta = \alpha$
 $\phi = \beta$

3)  TAKE, E.G., THE CIRCLE DEFINED BY

$\phi = 0$
 $\theta = \lambda$
 $r = R \cos \lambda$

$\phi = 0, \theta = \lambda, \frac{4\pi}{3}(r^3 - R^3 \cos^3 \lambda) = \int_0^t Q dt$

$\frac{4\pi}{3}(r^3 - R^3 \cos^3 \theta) = \int_0^t Q dt$

4)  $\theta = \theta_1$ → THIS EQ. DEFINES THE STREAMTUBE (A CONE OF SEMI-ANGLE θ_1)

5) $v_z = v_r \cos(\theta) = \frac{Q}{4\pi r^2} \cos\theta = \frac{Q z}{4\pi (x^2 + y^2 + z^2)^{3/2}}$

$v_x = v_r \sin\theta \cos\phi = \frac{Q}{4\pi r^2} \sin\theta \cos\phi = \frac{Q}{4\pi} \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$

$v_y = v_r \sin\theta \sin\phi = \frac{Q}{4\pi r^2} \sin\theta \sin\phi = \frac{Q}{4\pi} \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$

6) OBSERVER MOVING WITH VELOCITY $v_x = U, v_y = 0, v_z = 0$

$v'_x = \frac{Q}{4\pi} \frac{x}{(x^2 + y^2 + z^2)^{3/2}} - U$

$v'_y = \frac{Q}{4\pi} \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$

$v'_z = \frac{Q}{4\pi} \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$

... TO BE CONTINUED

STREAMLINES

$\frac{dz}{z} = \frac{dy}{y} = \frac{dx}{x} \Rightarrow \frac{x}{x_0} = \frac{y}{y_0} = \frac{z}{z_0}$

→ STRAIGHT LINES INTERSECTING THE ORIGIN

$\frac{dx}{dt} = \frac{Q}{4\pi} \frac{x}{(x^2 + y^2 + z^2)^{3/2}}$ $t=0$
 $x=x_0$
 $y=y_0$
 $z=z_0$

$\frac{dy}{dt} = \frac{Q}{4\pi} \frac{y}{(x^2 + y^2 + z^2)^{3/2}}$

$\frac{dz}{dt} = \frac{Q}{4\pi} \frac{z}{(x^2 + y^2 + z^2)^{3/2}}$

$\frac{x}{y} = \frac{x_0}{y_0}$
 $\frac{x}{z} = \frac{x_0}{z_0}$
 $\frac{y}{z} = \frac{y_0}{z_0}$

$\int_{x_0}^x x^2 dx = \int_0^t \frac{Q}{4\pi} \frac{1}{\left[1 + \left(\frac{y_0}{x_0}\right)^2 + \left(\frac{z_0}{x_0}\right)^2\right]^{3/2}} dt$